

Algebra 1  
Unit 9  
Lesson 4 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Organizers of a music festival charge \$60 for admission. The organizers polled the attendees to determine how increasing the price would affect attendance, and therefore affect profit. They discover that if they reduce the price by \$1, their profit would be maximized at \$600,000. However, if they raised the price by \$1, they would make \$560,000 in profit. The relationship between the music festival's profit,  $y$ , and the difference in the admission price,  $x$ , can be modeled by a quadratic equation.

1. Identify the vertex and another point from the context.
  - a. Vertex: (\_\_\_\_\_, \_\_\_\_\_)
  - b. Point: (\_\_\_\_\_, \_\_\_\_\_)
2. Find the value of  $a$  for the quadratic equation modeling the music festival's profit.
3. Write the quadratic equation that models the music festival's profit in vertex form.
4. Rewrite the equation in standard form.

A statistician for a running shoe company analyzes data from the years 2007 to 2014 to help inform the company about trends. The relationship between the number of people ages 6 to 17, in millions,  $y$ , who listed running as a hobby for  $x$  years since 2007 can be modeled by a quadratic equation. In 2011, 4 years into the study, the maximum number of runners in this age group was recorded at 10,000,000. Just one year earlier, there were 9,860,000 runners recorded in this age group.

5. Identify the vertex and another point from the context.
  - a. Vertex: (\_\_\_\_, \_\_\_\_\_)
  - b. Point: (\_\_\_\_, \_\_\_\_\_)
6. Find the value of  $a$  for the quadratic equation that models the relationship.
7. Write the quadratic equation that models the relationship in vertex form.

A quadratic equation can model the height above the ground,  $y$ , in feet, of a toy rocket after  $x$  seconds that it is launched from the top of a building. The rocket reaches a maximum height of 256 feet above the ground after 2 seconds of flight time. The rocket hits the ground 6 seconds after its launch.

8. Identify the vertex and another point from the context.
  - a. Vertex: (\_\_\_\_, \_\_\_\_\_)
  - b. Point: (\_\_\_\_, \_\_\_\_\_)
9. Find the value of  $a$  for the quadratic equation modeling the rocket's height.
10. Write the quadratic equation that models the rocket's height in vertex form.